How-To Library

Informatica

Prerequisites to use service principal authentication

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Abstract

This article explains the prerequisite tasks that you must complete before you configure service principal authentication to connect to Microsoft Azure Synapse SQL.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector

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Overview

You can use Microsoft Azure Synapse SQL Connector to connect to Microsoft Azure Synapse SQL using service principal authentication.

Before you create a Microsoft Azure Synapse SQL connection using service principal authentication, complete the following prerequisite tasks:

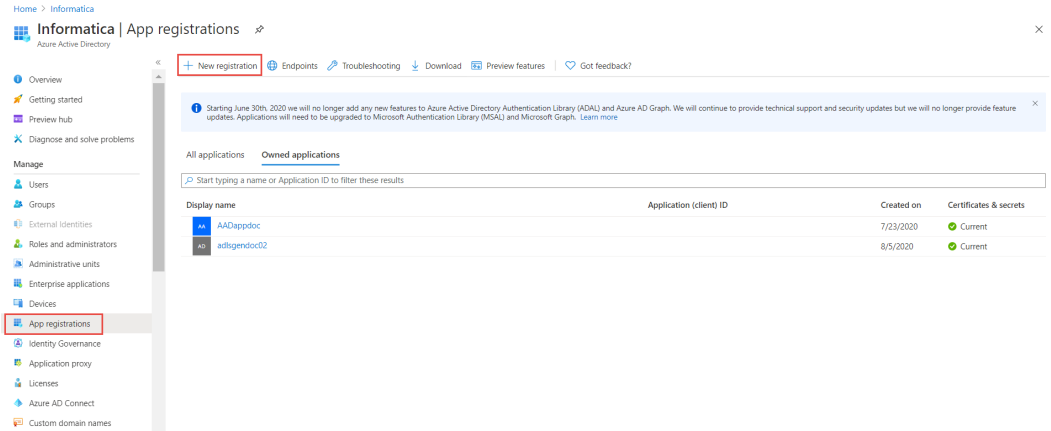
1. Register a service principal application.
2. Configure a service principal user for serverless SQL pool.
3. Configure a service principal user for dedicated SQL pool.

For information about configuring a Microsoft Azure Synapse SQL connection, see the *Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector documentation*.

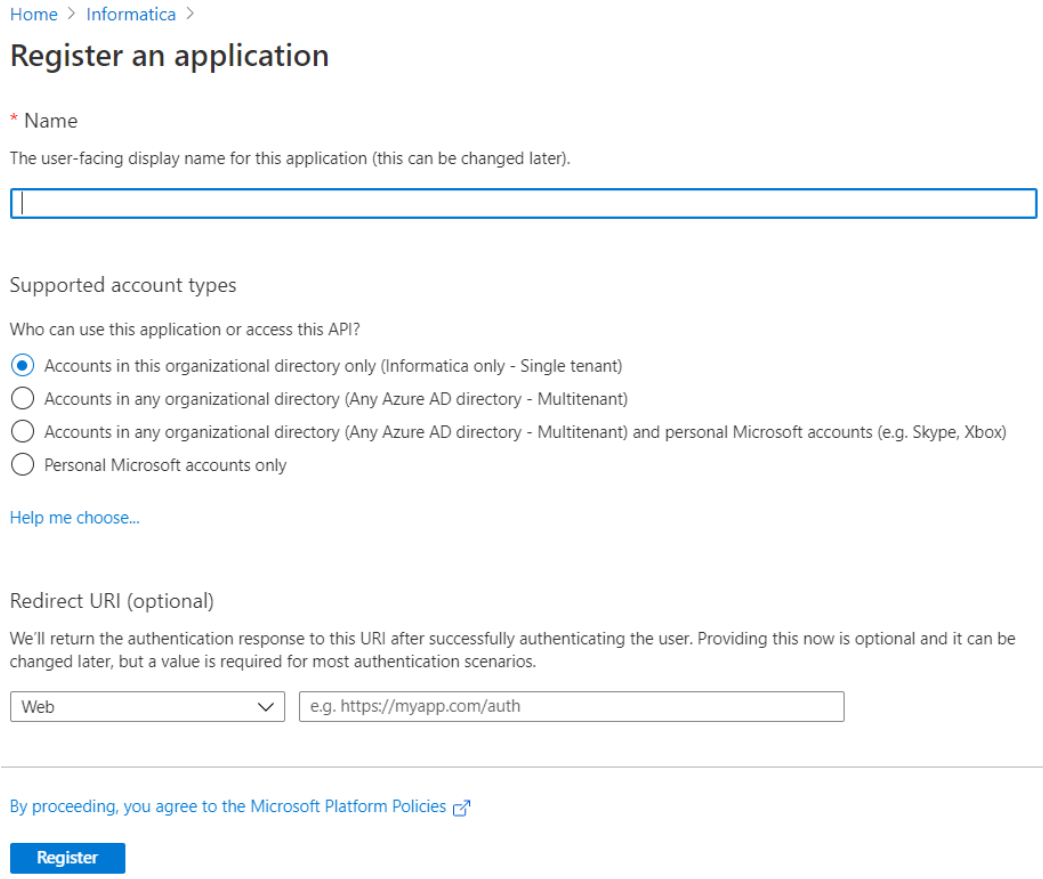
Registering a service principal application

Register a service principal application in the Azure portal.

1. Log in to the following Azure portal: https://portal.azure.com/
2. Navigate to **App registrations**.

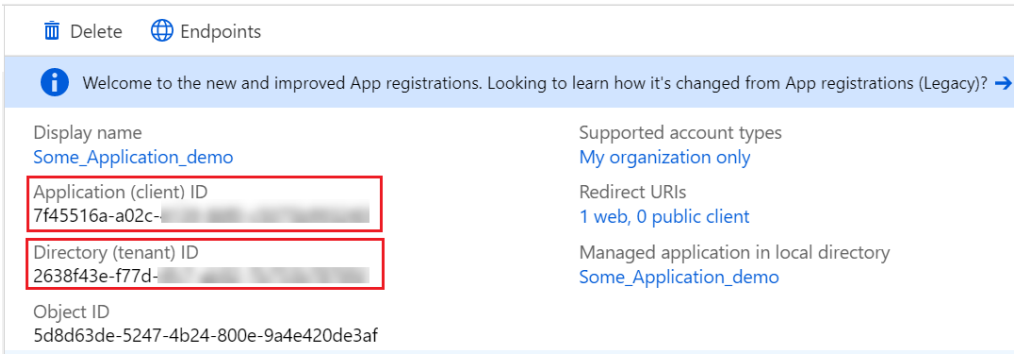


3. Click **New registration** to create a new service principal application.
4. On the **Register an application** page, enter the details for the new application.



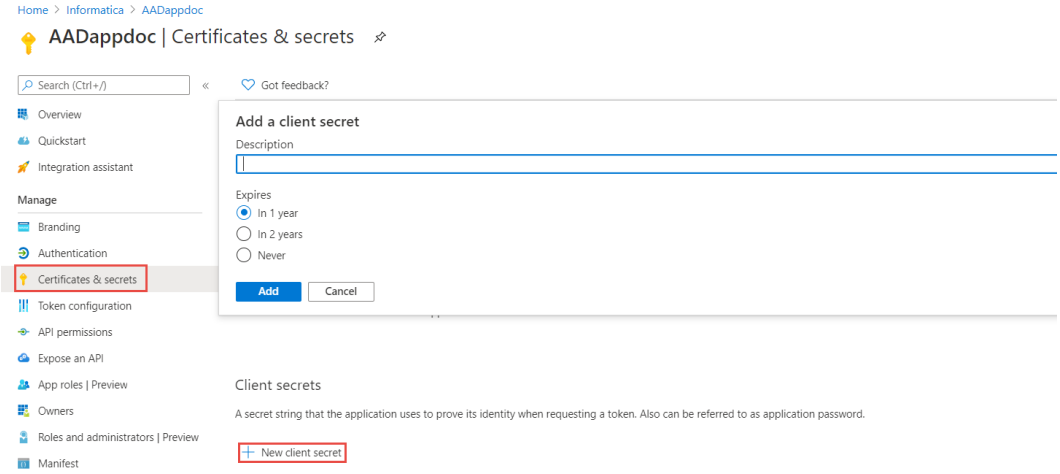
   a. In the **Name** field, enter the application name.
   b. In the **Redirect URI** section, select **Web** as the type of the application and enter the URL of the application.
   c. Click **Register**.

      The details of the newly created Service Principal application page are displayed.

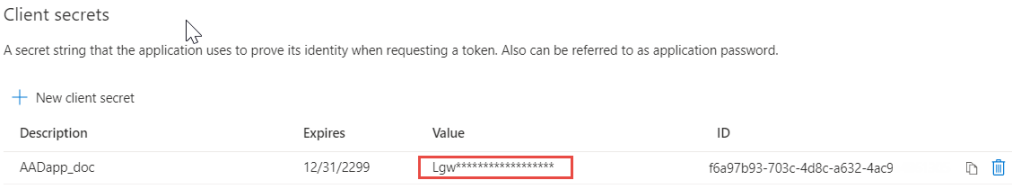


5. In the Manage section, click **Certificates & secrets** section.

6. Click **New client secret**.



7. In the **Add a client secret** page, perform the following steps:
   a. Enter a name for the client secret in the **Description** field.
   b. In the **Expires** field, you can select the duration of the key as **Never**.
   c. Click **Add**.
   d. The value of the key is generated and displayed in the **Value** field.



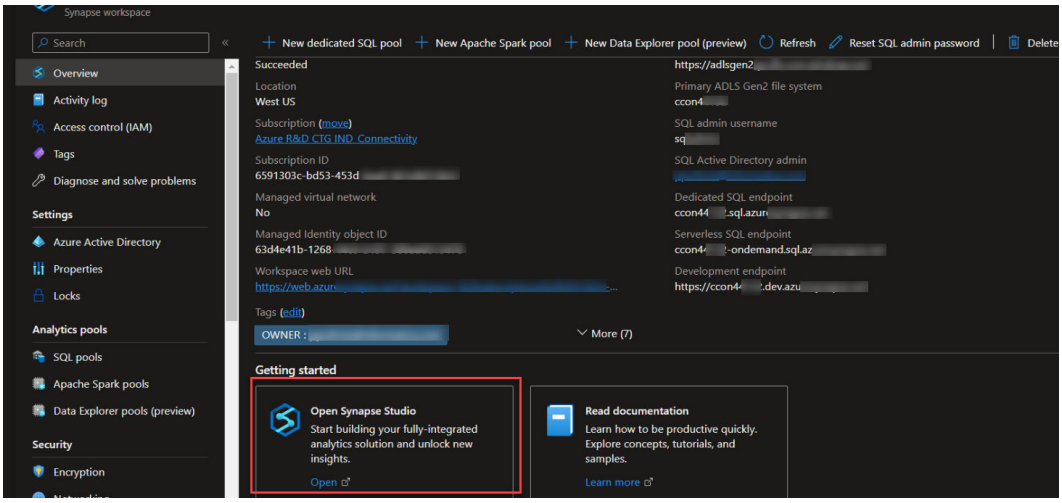
Note: You must copy the key value as you cannot retrieve the value once you leave the page. Ensure that the client secret does not contain special characters.

Configuring service principal user for serverless SQL pool

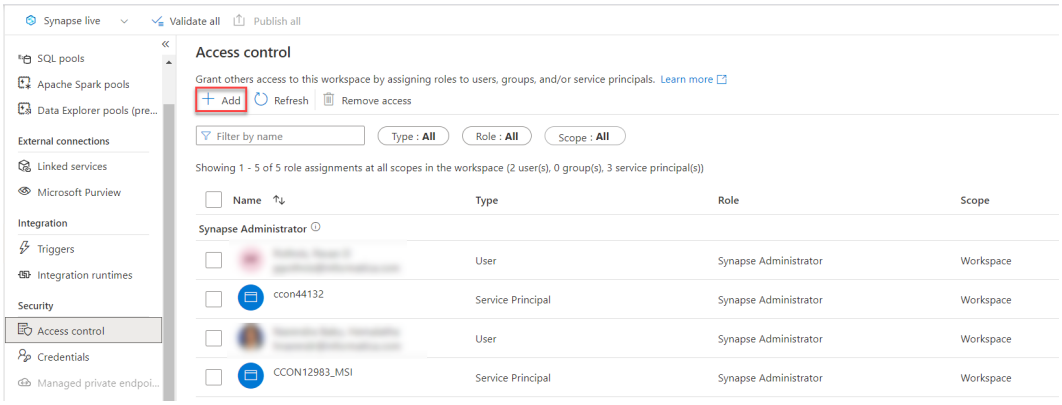
Create a service principal user for the serverless SQL pool and assign the Storage Blob Data Contributor role.

1. On the Azure portal, navigate to the serverless SQL pool.

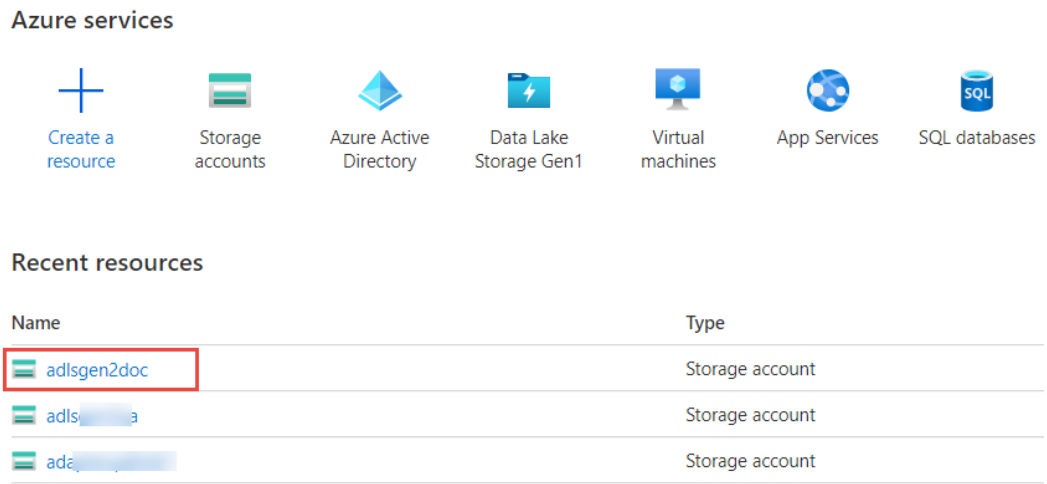
2. Access **Open Synapse Studio**.



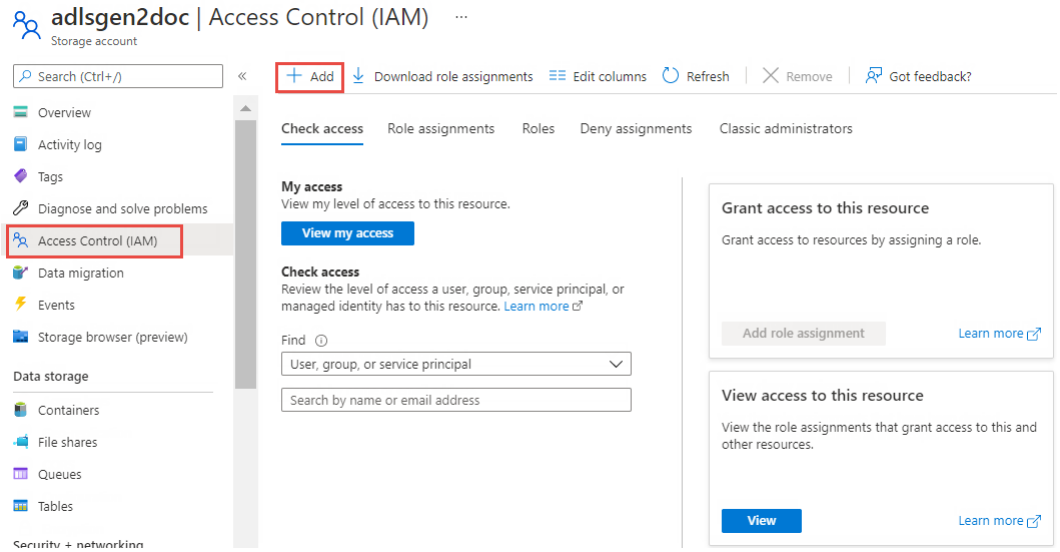
3. In the **Manage** section, click **Access control**.



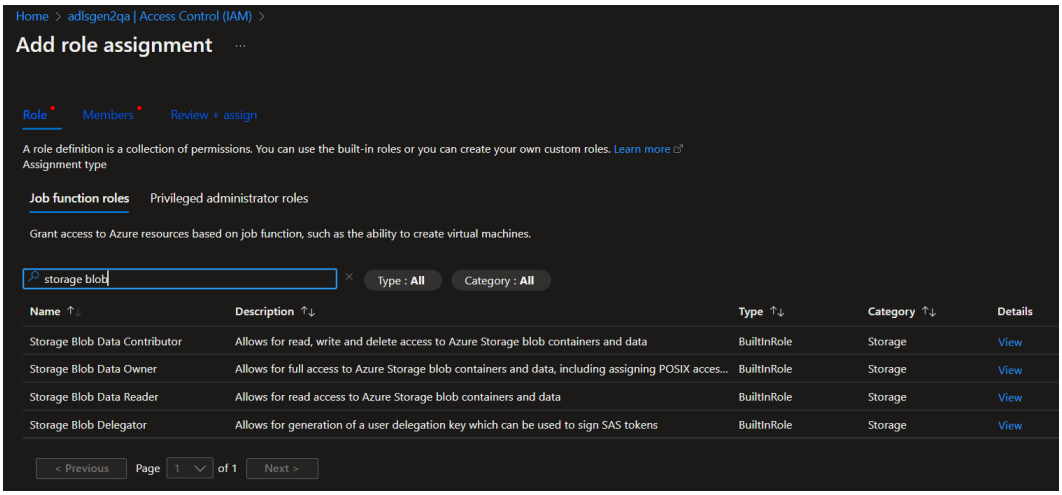
4. Click **Add** to add the service principal application.
5. Navigate to the Microsoft Azure Data Lake Storage Gen2 storage account.



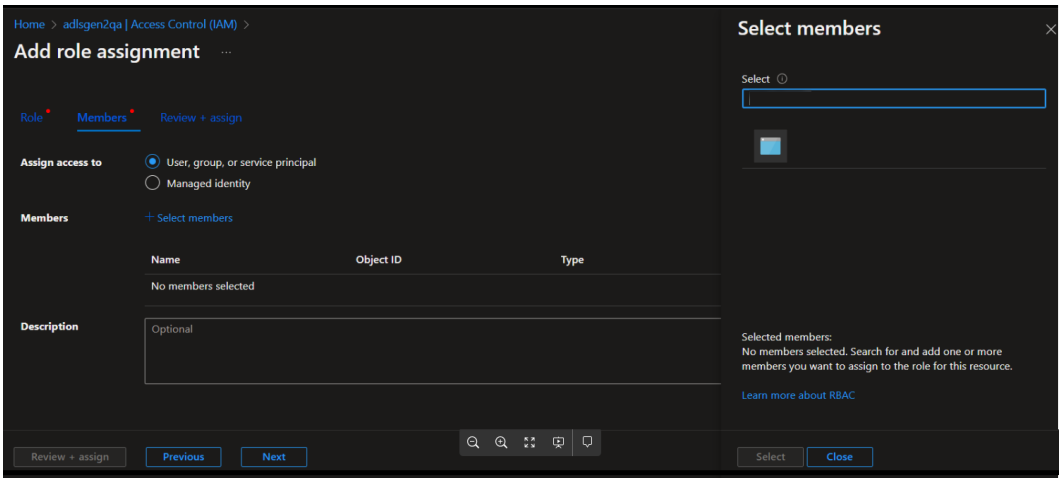
6. Click **Access control (IAM)** > **Add**.



7. In the **Add role assignment** page, select the **Role** tab, search for **Storage Blob Data Contributor** role, and then click **Next**.



8. On the **Members** tab, click **Select members** and search for the service principal application created earlier.



9. Select the service principal application and assign the role.

Configuring service principal user for dedicated SQL pool

Create a service principal user for the dedicated SQL pool and assign the Storage Blob Data Contributor role.

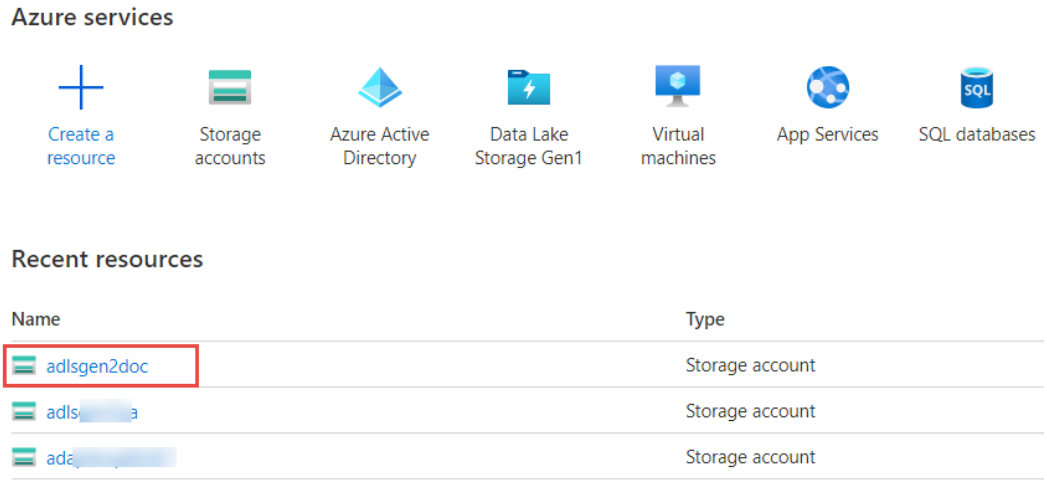
1. Access Microsoft SQL Server Management Studio, Azure Data Studio, or any other tool that allows access to Azure database and run the following commands:

```
USE SQLDW_DEV
CREATE USER [servprincipalapp] FROM EXTERNAL PROVIDER
grant ALTER ANY EXTERNAL FILE FORMAT to [serviceprincipalapp];
grant ALTER ANY EXTERNAL DATA SOURCE to [serviceprincipalapp];
grant create table to [serviceprincipalapp];
grant create schema to [serviceprincipalapp];
grant select to [serviceprincipalapp];
grant update to [serviceprincipalapp];
grant insert to [serviceprincipalapp];
grant delete to [serviceprincipalapp];
grant create view to [serviceprincipalapp];
grant select on schema :: sys to [serviceprincipalapp]
grant control to [serviceprincipalapp]
or
grant ALTER ANY SCHEMA TO <servprincipalapp>;// To grant permissions only on the schema.
grant CREATE TABLE TO <servprincipalapp>;
```

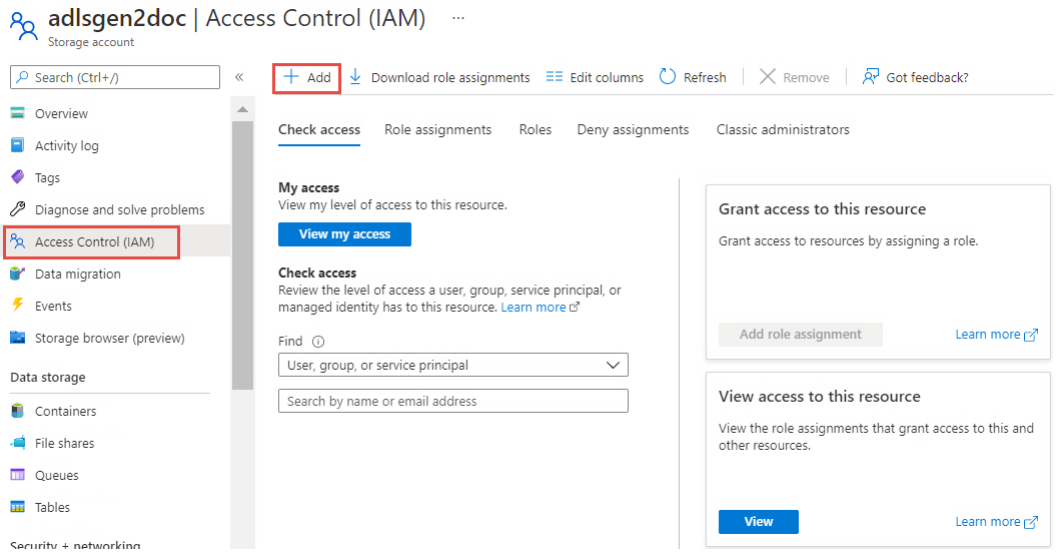
where, `SQLDW_DEV` is the name of the database and `serviceprincipalapp` is the service principal application. You can also create the user in master database. Ensure that the service principal application name matches the one that was configured in the previous steps.
If you have the ALTER ANY SCHEMA permissions, you must create the Master Key, Database Scoped Credential, and External Data Source in Microsoft Azure Synapse SQL that require the CONTROL permission on the database and specify the external data source when you create a connection.

Also, Microsoft Azure Synapse SQL Connector does not delete the Database Scoped Credential and External Data Source. You must manually delete the Database Scoped Credential and External Data Source.

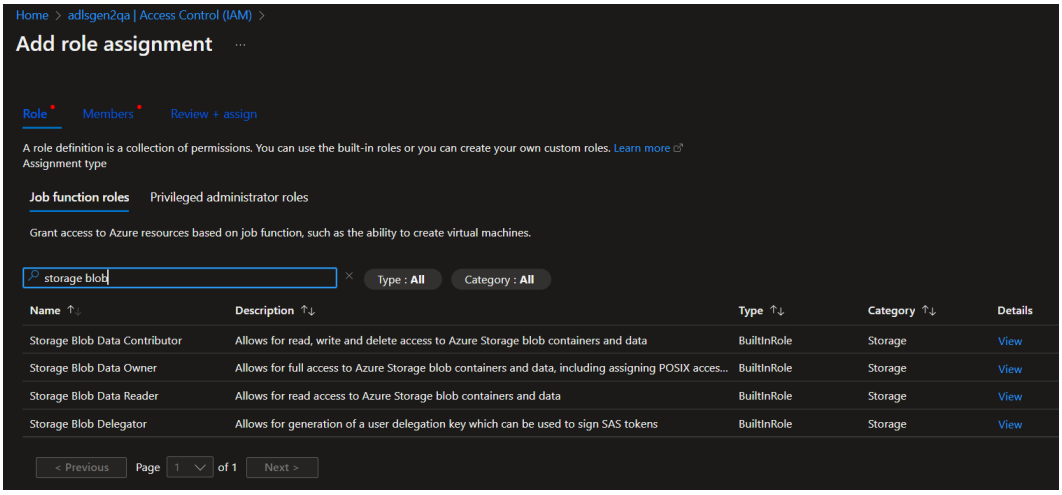
2. Navigate to the Microsoft Azure Data Lake Storage Gen2 storage account.



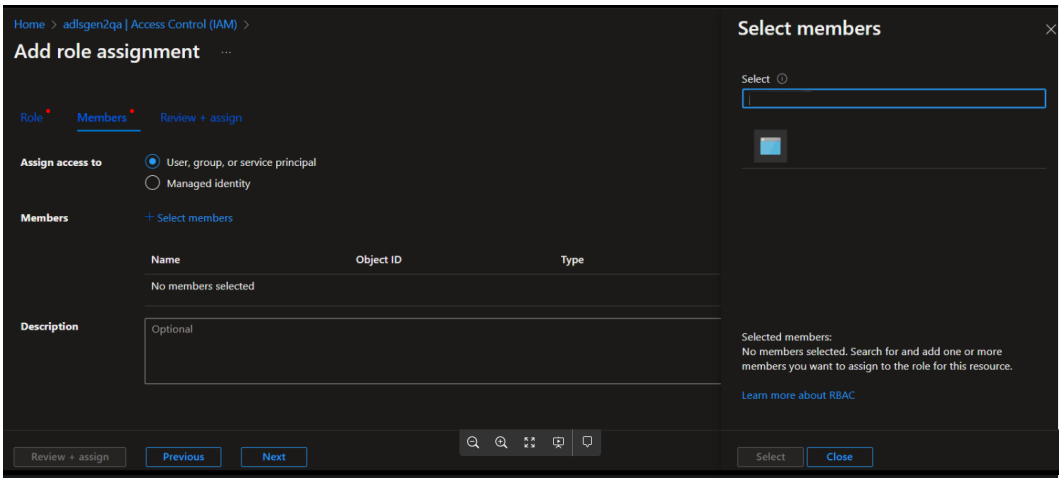
3. Click **Access control (IAM)** > **Add**.



4. In the **Add role assignment** page, select the **Role** tab, search for **Storage Blob Data Contributor** role, and then click **Next**.



5. On the **Members** tab, click **Select members** and search for the service principal application created earlier.



6. Select the service principal application and assign the role.

Author

Radha Viswanath

Acknowledgements

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